

Cohomological Asymptotics for linear Gps

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October 6, 2026

~ Summary ~

§1 Group cohomology...

§2 ... is just linear algebra, really!

§3 From $\mathbb{C}$ to $\mathbb{Q}_p$

§4 Iwasawa algebras

§1. Group cohomology.

Let Γ be a group.

$$\rho: \Gamma \rightarrow GL(V), \quad V \simeq \mathbb{C}^n$$

Question How does $H^i(\Gamma, V)$ grows as a function of p ?

⚠ $\dim_{\mathbb{C}} H^i(\Gamma, V)$ could be infinite.

Definition Γ is of type $FP_{\infty}(\mathbb{C})$ if $\exists$ resolution

$$\cdots \rightarrow \mathbb{C}\Gamma^{a_3} \rightarrow \mathbb{C}\Gamma^{a_2} \rightarrow \mathbb{C}\Gamma^{a_1} \rightarrow \mathbb{C}\Gamma^{a_0} \rightarrow \mathbb{C} \rightarrow 0$$

Facts (i) If Γ is $FP_{\infty}(\mathbb{C})$, $\dim_{\mathbb{C}} H^i(\Gamma, V) < \infty$.

(ii) The following Γ are $FP_\infty(\mathbb{C})$:

- finite Γ
- $\Gamma = \pi_1(M)$, M compact manifold
- Γ lattice in a Lie group.

(iii) If Γ is $FP_\infty(\mathbb{C})$, $\exists M = M(\Gamma, i)$ s.t.

$$\dim_{\mathbb{C}} H^i(\Gamma, V) \leq M \cdot \dim_{\mathbb{C}} V$$

$$\forall \rho: \Gamma \rightarrow GL(V).$$

Assume $\Gamma \leq_{f.g.} GL_n(\mathbb{C})$ & Γ is $FP_\infty(\mathbb{C})$.

Then, Γ is residually finite: (Mal'cev, 1940)

$\exists \Gamma \gg \Gamma_1 \gg \Gamma_2 \gg \dots$, each $\Gamma_i \trianglelefteq_{f.i.} \Gamma$ with $\bigcap_i \Gamma_i = \{1\}$.

$$\text{Let } V_i = \mathbb{C}[\Gamma/\Gamma_i].$$

Remark $\dim_{\mathbb{C}} V_i = (\Gamma : \Gamma_i)$ & $H^j(\Gamma, V_i) \cong H^j(\Gamma_i, \mathbb{C})$.

Theorem (W. Lück, '94) For any such chain Γ_i ,

$$\lim_{i \rightarrow \infty} \frac{\dim_{\mathbb{C}} H^j(\Gamma, V_i)}{(\Gamma : \Gamma_i)} = b_j^{(2)}(\Gamma)$$

↗ L^2 -Betti numbers.

Examples (i) Γ finite, $b_j^{(2)}(\Gamma) = \begin{cases} 1/|\Gamma|, & j=0 \\ 0, & j>0. \end{cases}$

(ii) Γ free of rank n , $b_j^{(2)}(\Gamma) = \begin{cases} n-1, & j=1 \\ 0, & j \neq 1. \end{cases}$

(iii) $\Gamma = \pi_1(S_g)$, $b_j^{(2)}(\Gamma) = \begin{cases} g-2, & j=1 \\ 0, & j \neq 1. \end{cases}$

(iv) (W. Lück, J. Lott, 1995) M hyperbolic, finite volume 3-manifold,
 $\Gamma = \pi_1(M)$, $b_j^{(2)} = 0 \quad \forall j$.

Assume $G \leq GL_n \mathbb{C}$ is a semi-simple complex lie group
 and moreover $\Gamma \leq G$.

If $\rho: G \rightarrow GL(V)$ is a lie rep, then usually V and
 $\mathbb{C}[\Gamma]$ are very distinct.

Fact There exists $k = \text{rank } G \in \mathbb{Z}_{\geq 0}$ s.t.

$$V \cong V_{(\lambda_1, \dots, \lambda_k)} \quad , \quad (\lambda_1, \dots, \lambda_k) \in \mathbb{Z}_{\geq 0}^k$$

whenever $G \rightarrow GL(V)$ is irreducible.

Examples (i) $G \cong SL_2 \mathbb{C}$, $k=1$, $V_{\lambda_1} = \text{Sym}^{\lambda_1}(\mathbb{C}^2)$
 $= \mathbb{C}[x, y]_{\lambda_1}$.
 $\uparrow$ hom. poly. of degree λ_1 .

(ii) $G \cong PSL_2 \mathbb{C}$, $V \cong V_{2\lambda}$, $\leftarrow$ weight must be even.

(iii) $G \cong SL_3 \mathbb{C}$, $k=2$, $V_{(\lambda_1, 0)} = \text{Sym}^{\lambda_1}(\mathbb{C}^3)$,
 $V_{(0, \lambda_2)} = \text{Sym}^{\lambda_2}(\mathbb{C}^3)^*$
 $V_{(\lambda_1, \lambda_2)} \hookrightarrow \mathbb{C}[\text{Flag variety}]$.

Conjecture (L. Guerrero, S. 2024) Assume $\Gamma \cap Z(G) = \{1\}$.

Then

$$\frac{\dim_{\mathbb{C}} H^j(\Gamma, V)}{\dim_{\mathbb{C}} V} \rightarrow b_j^{(2)}(\Gamma) \quad (*)$$

as $\min \{ \lambda_1, \dots, \lambda_k \} \rightarrow \infty$.

(building on W. Fu, 2024)

Theorem (Guerrero-S., 2024) (*) holds for $G = (P)SL_2 \mathbb{C}$.

Theorem (S., in preparation) (*) holds for $G = (P)SL_3 \mathbb{C}$.
and $V = V(t, t)$.

§2. ... is just linear algebra, really!

Write

$$\dots \rightarrow \mathbb{C}\Gamma^{a_3} \rightarrow \mathbb{C}\Gamma^{a_2} \rightarrow \mathbb{C}\Gamma^{a_1} \rightarrow \mathbb{C}\Gamma^{a_0} \rightarrow \mathbb{C} \rightarrow 0$$

There are matrices $A \in \text{Mat}_{a_j \times a_{j+1}} \mathbb{C}\Gamma$, $B \in \text{Mat}_{a_{j-1} \times a_j} \mathbb{C}\Gamma$ s.t.

$$\dots \rightarrow \mathbb{C}\Gamma^{a_{j+1}} \xrightarrow{A \cdot} \mathbb{C}\Gamma^{a_j} \xrightarrow{B \cdot} \mathbb{C}\Gamma^{a_{j-1}} \rightarrow \dots$$

For any $\Gamma \rightarrow GL(V)$ with $\dim V = m$, they define:

$$A_V \in \text{Mat}_{a_j m \times a_{j+1} m} \mathbb{C}, \quad B_V \in \text{Mat}_{a_{j-1} m \times a_j m} \mathbb{C}$$

s.t.

(The most important) Fact.

$$\dim_{\mathbb{C}} H^i(\Gamma, V) = a_j m - \text{rk}_{\mathbb{C}} A_V - \text{rk}_{\mathbb{C}} B_V.$$

Indeed,

$$H^i(\Gamma, V) = \frac{\ker \left(V^{a_j} \xrightarrow{B_V^T} V^{a_{j+1}} \right)}{\text{Im} \left(V^{a_{j-1}} \xrightarrow{A_V^T} V^{a_j} \right)}.$$



Definition Let R be a ring with 1. A Sylvester rank function rk is a map $\text{rk}: \text{Mat}(R) \rightarrow \mathbb{R}_{\geq 0}$ s.t.

$$(1) \text{rk}(0) = 0, \text{rk}(1) = 1.$$

$$(2) \text{rk}(AB) \leq \min\{\text{rk}(A), \text{rk}(B)\}$$

$$(3) \text{rk}\begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix} = \text{rk}(A) + \text{rk}(B)$$

$$(4) \text{rk}\begin{pmatrix} A & C \\ 0 & B \end{pmatrix} \geq \text{rk}(A) + \text{rk}(B).$$

(division ring)

Examples (i) Over a field F , $\text{rk} = \text{rk}_F \quad \forall \text{rk}.$

(ii) if $\rho: \Gamma \rightarrow \text{GL}(V)$, then

$$\text{rk}_\rho(A) = \frac{\text{rk}_\mathbb{C} \rho(A)}{\dim_\mathbb{C} V}$$

is a Sylvester rk fm on $\mathbb{C}$.

(iii) Taking $V_i = \mathbb{C}[\Gamma/\Gamma_i]$, $\text{rk}_\Gamma = \lim_{i \rightarrow \infty} \text{rk}_{V_i}$ is a Sylvester rank fm. We have

$$b_j^{(2)}(\Gamma) = (a_j - \text{rk}_\Gamma(A) - \text{rk}_\Gamma(B)).$$

(A. Jaikin-Zapirain, 2019)

(iv) If $\phi: R \rightarrow S$ is a ring map is rk is a Sylvester rank fm on S , then $(\phi^* \text{rk})(A) = \text{rk}(\phi(A))$ is a Sylvester rank fm on R . (ii) & (iii) are particular cases of this.

Corollary To prove (*), it suffices to show

$$\text{rk}_V \rightarrow \text{rk}_\Gamma \quad (**)$$

as $\min\{\lambda_1, \dots, \lambda_k\} \rightarrow \infty.$

§ 3. From $\mathbb{C}$ to $\mathbb{Q}_p$

Let $A \in \text{Mat}(\mathbb{C}\Gamma)$. Then, $\exists S \leq \mathbb{C}$ f.g. subring s.t.

- $\Gamma \leq G(S)$ $A \in \text{Mat}(S\Gamma)$.
 $\uparrow$ S -points, G seen as split algebraic gp over $\mathbb{Z}$

Theorem (Cassels, '76) For infinitely many primes p ,
 $\exists$ embedding $S \hookrightarrow \mathbb{Z}_p$,
 $\uparrow$ p -adic integers.

Thus:

- $\Gamma \leq G(\mathbb{Z}_p)$, $A \in \text{Mat}(\mathbb{Z}_p\Gamma)$.

(Not very easy) fact. $(*)$ passes to finite index overgroups.
 So we may assume $\Gamma \leq G^1(\mathbb{Z}_p) = \ker(G(\mathbb{Z}_p) \rightarrow G(\mathbb{F}_p))$

Let $\phi: S\Gamma \hookrightarrow \mathbb{Q}_p G^1$. All the reps. $V = V_{(\lambda_1, \dots, \lambda_k)}$ are defined over $\mathbb{Z}$, so $\exists W = W_{(\lambda_1, \dots, \lambda_k)}$ over $\mathbb{Q}_p$ and $G^1 \rightarrow GL(W)$ s.t. $\Gamma \kappa_V(A) = (\phi^* \Gamma \kappa_W)(A) = \Gamma \kappa_W(\phi(A))$.
 What about $\Gamma \kappa_\Gamma$?

Theorem (M. Lazard, '65) There exists a division ring $\mathcal{D}$ and an embedding $\mathbb{Q}_p G^1 \hookrightarrow \mathcal{D}$.

Theorem (M. Harris, '79) + Jacquin-Lück approx. We have
 $\Gamma \kappa_\Gamma A = (\phi^* \Gamma \kappa_{\mathcal{D}})(A) = \Gamma \kappa_{\mathcal{D}} \phi(A)$.

Hence, it suffices to prove

$$\Gamma \kappa_W \rightarrow \Gamma \kappa_{\mathcal{D}} \quad (***)$$

(for all but finitely many p)

on $\mathbb{Q}_p G^1$ as $\min\{\lambda_1, \dots, \lambda_k\} \rightarrow \infty$.

(Fairly easy but obscure) Fact. To show (***), it suffices to prove

$$\frac{\dim_{\mathbb{Q}_p} {}^x W}{\dim_{\mathbb{Q}_p} W} \rightarrow 1 \quad (***) \text{ bis}$$

as $\min \{\lambda_1, \dots, \lambda_k\} \rightarrow \infty$.

§4 Iwasawa algebras

Let $G^m = \ker(G(\mathbb{Z}_p) \rightarrow G(\mathbb{Z}/p^m\mathbb{Z}))$. Then

$$G^\mathbb{Z} = \varprojlim G^\mathbb{Z}/G^m$$

is a profinite (even pro- p) group.

There is a profinite ring

$$\mathbb{Z}_p[[G^\mathbb{Z}]] = \varprojlim \mathbb{Z}_p[G^\mathbb{Z}/G^m],$$

and $\mathbb{Q}_p G^\mathbb{Z} \hookrightarrow \mathbb{Q}_p\{\{G^\mathbb{Z}\}\} := \mathbb{Q}_p \otimes_{\mathbb{Z}_p} \mathbb{Z}_p[[G^\mathbb{Z}]]$, the Iwasawa algebra of $G^\mathbb{Z}$.

Assume $p \neq 2$, and using $G^\mathbb{Z} \leq GL_n(\mathbb{Z}_p)$ consider the function

$$\log: G^\mathbb{Z} \rightarrow {}_p\text{Mat}(\mathbb{Z}_p).$$

$\mathfrak{g}_0 = \frac{1}{p} \log(G^\mathbb{Z})$ is a Lie algebra over $\mathbb{Z}_p$, free of rank $\dim G$ as a $\mathbb{Z}_p$ -module.

Take $U(g_0) = \text{UEA}$ over $\mathbb{Z}_p$,

$$\widehat{U}(g_0) = \varprojlim U(g_0)/p^n U(g_0)$$

$$\widehat{U} = \mathbb{Q}_p \otimes_{\mathbb{Z}_p} \widehat{U}(g_0)$$

Then,

$$\exp: g_0 \longrightarrow \widehat{U}^\times$$

converges. Since $G^\perp \xrightarrow{\log} g_0 \xrightarrow{\exp} \widehat{U}^\times$ is a continuous gp hom, we have a map:

$$\mathbb{Q}_p\{G^\perp\} \longrightarrow \widehat{U}.$$

Theorem (K. Ardakov, S. Wodzley, 2013) $\mathbb{Q}_p\{G^\perp\} \rightarrow \widehat{U}$ is injective.

(Nice) Facts (i) $\widehat{U}$ is easier to describe than $\mathbb{Q}_p G^\perp$. Elements are power series in a $\mathbb{Z}_p$ -basis of g_0 .

(ii) The action $\mathbb{Q}_p G^\perp \curvearrowright W$ extends to $\widehat{U} \curvearrowright W$.

(iii) It is false that

$$\frac{\dim_{\mathbb{Q}_p} xW}{\dim_{\mathbb{Q}_p} W} \rightarrow 1$$

for arbitrary $x \in \widehat{U}$. (Minimal exp. $g_0 = \mathfrak{sl}_2 \mathbb{Z}_p \times \mathfrak{sl}_2 \mathbb{Z}_p$, $x \in \mathbb{Z}(\widehat{U})$).

Theorem (Ardakov - Wadsley, 2014) Let $\chi_W: Z(\hat{U}) \rightarrow \mathbb{Q}_p$ be the central character of $\hat{U} \curvearrowright \mathbb{Q}_p$. For all but finitely many p , the composition

$$\mathbb{Q}_p\{\{G^1\}\} \rightarrow \hat{U} \rightarrow \hat{U}/\hat{U}(\ker \chi_W)$$

is injective for all W .

Definition $x \in \hat{U}$ is generic if $x \notin \hat{U}(\ker \chi_W) \forall W$.

Theorem (W. Fu, 2024) If $g_0 = (\mathfrak{sl}_2 \mathbb{Z}_p)^{\kappa}$ and x is generic, then

$$\frac{\dim_{\mathbb{Q}_p} xW}{\dim_{\mathbb{Q}_p} W} \rightarrow 1$$

as $\min\{\lambda_1, \dots, \lambda_k\} \rightarrow \infty$

Theorem (S., in prep.) If $g_0 = (\mathfrak{sl}_2 \mathbb{Z}_p)^a \times (\mathfrak{sl}_3 \mathbb{Z}_p)^b$ and x is generic, then

$$\frac{\dim_{\mathbb{Q}_p} xW}{\dim_{\mathbb{Q}_p} W} \rightarrow 1$$

as $\min\{\lambda_1, \dots, \lambda_k\} \rightarrow \infty$.